

Pan Ziyue



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EDUCATION

Singapore University of Technology and Design

Aug 2022 — May 2026

Bachelor of Engineering, Engineering Product Development, Double Minor in Computer Science and Digital Humanities

- Cumulative GPA: 4.2/5.0 | Global Distinguished Scholarship

Singapore Polytechnic

Apr 2016 — May 2019

Diploma with Merit in Computer Engineering

- Cumulative GPA: 3.965/4.0 | Diploma Silver Medalist, Director's Honour Roll AY2016/17, AY2017/18

EXPERIENCE

Embedded Systems Intern

Aug 2024 — Jan 2025

Equatorial Space Systems Pte. Ltd.

- Designed a PC/104-compatible pyro deployment board for actuating high-current rocket deployment events
- Developed a Python-based Software-in-the-Loop test software to replay historical flight data as simulated sensor inputs
- Implemented and tested key drivers in C++, adhering to established high reliability code standards (e.g. MISRA C++)

Cofounder

May 2013 — Present

FourierIndustries (Personal Startup)

- Developed in-house mobile applications (2013–2015), including a project awarded the 2014 Apple WWDC Scholarship
- Designed and produced multiple generations of IoT microcontroller boards (SPEEduino, SSTuino, SSTuino II)
- Authored teaching materials for the SSTuino II educational IoT microcontroller board
- Led schematic design, PCB layout, firmware development, and small-batch manufacturing
- Provisioned, maintained and operated FDM 3D printers for 3D printing services provided by the company

Research Engineering Intern

Oct 2018 — Mar 2019

Earth Observatory of Singapore, Nanyang Technological University

- Assisted in the consolidation and stabilization of the Microsoft Azure port of the Hybrid Science Data System (HySDS), working in collaboration with NASA's Jet Propulsion Laboratory (JPL) engineers
- Wrote Terraform, shell and Python scripts to automate deployment on Azure VMs, blob storage and network infrastructure
- Benchmarked cost-performance of the Azure port versus the original AWS version and realized a cost saving of up to 50%

Intern

Mar 2018 — Apr 2018

Satellite Research Centre, Nanyang Technological University

- Developed ORBITAL, a Java desktop application to help highlight anomalies in satellite orbital data
- Authored a proposal and feasibility evaluation for HopeSat-1, a 1U educational CubeSat

Intern

Mar 2016 — Apr 2016

buUuk Pte. Ltd.

- Aided in the development of a client iOS mobile application project in Objective-C

PROJECTS

Team Lead, ESS-SUTD Rocket Flight Computer System (University+Industry Capstone) (<https://capstoneshowcase.sutd.edu.sg/project/proj-s10-equatorial-flight-controller-for-rocketry/>)

Sep 2025 — Apr 2026

- Capstone project with Equatorial Space Systems to develop a COTS-based, fault-tolerant flight controller system targeting orbital-class launch vehicles at a fraction of traditional avionics cost
 - Led architectural design of a dual-redundant flight computer system with dual-CAN backbone, drawing on heritage architectures
 - Designed and fabricated custom PC/104-format PCBs (OBC, ACU, Ethernet switch boards) with latchup-protected power chains and MIL-spec connectors

Electronics Lead, Electric Vehicle by Additive Manufacturing (EVAM 1.0)

Nov 2024 — Jan 2026

- EVAM 1.0 is the flagship project of the Electric Vehicle club in SUTD, which aims to build a completely functional, full scale electric vehicle for motorsports

- Built and tested a redesigned vehicle electronics system, with the main aim of improving data transfer reliability
- Mentored fellow students on the team on electronics CAD design and embedded platform programming

Avionics Team Member, Project Mynah (SEDS Singapore Rocketry Team) Sep 2022 — Jun 2023

- Project Mynah is the first amateur rocket built by students from various Singaporean universities
 - Built, integrated and tested commercially available flight computers, including firmware and radio systems
 - Supported avionics readiness for Friends of Amateur Rocketry (FAR) 2023, with the team attaining 3rd Place

Team Member, Non-industrial Decomposition of PLA Plastics May 2023 — Sept 2023

- Undergraduate Research Project to investigate the decomposition of PLA plastic, a common plastic used in 3D printing, under non-industrial settings
 - Developed embedded RP2040 CircuitPython code for Internet-based data logging and Jupyter notebook based code for experimental data analysis.
 - Sun tracking analysis was completed using AGI Systems Tool Kit.

Team Member, Airbus-SSTA HADR Challenge May 2018 — Sept 2018

- Winner of the 2018 Airbus-SSTA Humanitarian Aid and Disaster Relief Challenge, organised by Airbus Defence & Space in collaboration with the Singapore Space and Technology Association.
 - Tasked with developing an Android application that increases the efficiency and effectiveness of Humanitarian Assistance and Disaster Relief operations

PUBLICATIONS

- Teo, T. H., Ziyue, P., Wesley, N. A. H., Htet, K. K., Subong, K. C., Lam, H. S., & Xiang, M. (2025, May 1). Cryptographically Secure Random Number Generator utilizing Environmental Radiation. *TENCON 2025 - 2025 IEEE Region 10 Conference (TENCON)*. <https://doi.org/10.1109/TENCON66050.2025.11375162>

SKILLS

- **Software Frameworks and Languages:** iOS SDK (Swift, Objective-C), Android SDK (Kotlin), ROS, Linux (RHEL, Debian), Docker, NodeJS, Svelte, jQuery, TypeScript, Terraform, PHP 7
- **Embedded Platforms:** AVR (Arduino C++), RP2040 (MicroPython), ESP32 (ESP-IDF, FreeRTOS, C), ESP8266 (Mongoose OS, C), Verilog (HDL), Lucid (HDL)
- **Computer Aided Design Tools:** KiCad, Autodesk EAGLE, Autodesk Fusion 360
- **Certifications:** AGI Systems Tool Kit (Level 2 certified), [FCC Amateur Radio Technician License](#)
- **Fabrication:** FDM+SLA 3D Printing, Laser Cutter, LDKF Printed Circuit Board CNC Router, SMD Manual Soldering, Manual Lathe